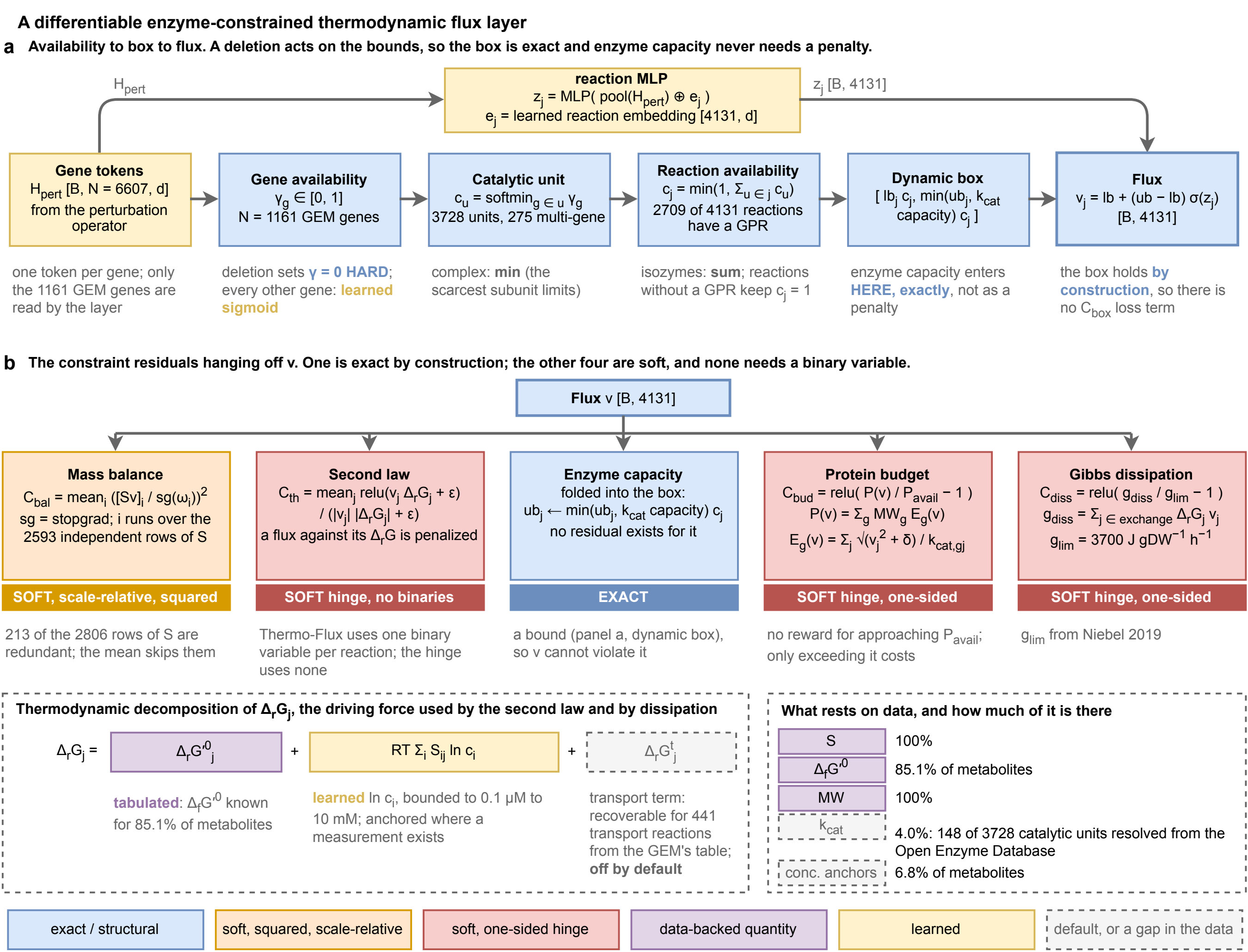

A differentiable enzyme-constrained thermodynamic flux layer
a Availability to box to flux. A deletion acts on the bounds, so the box is exact and enzyme capacity never needs a penalty.
H_{pert}
reaction MLP
z_j = MLP(pool(H_{pert}) ⊕ e_j)
e_j = learned reaction embedding [4131, d]
z_j [B, 4131]
Gene tokens
H_{pert} [B, N = 6607, d]
from the perturbation operator
Gene availability
$\gamma_g \in [0, 1]$
N = 1161 GEM genes
Catalytic unit
$c_u = \mathrm{softmin}_{g \in u}\, \gamma_g$
3728 units, 275 multi-gene
Reaction availability
$c_j = \min(1, \Sigma_{u \in j}\, c_u)$
2709 of 4131 reactions have a GPR
Dynamic box
[$lb_j\, c_j$, min(ub_j, k_{cat} capacity) c_j]
Flux
$v_j = lb + (ub - lb)\, \sigma(z_j)$
[B, 4131]
one token per gene; only the 1161 GEM genes are read by the layer
deletion sets γ = 0 HARD; every other gene: learned sigmoid
complex: min (the scarcest subunit limits)
isozymes: sum; reactions without a GPR keep c_j = 1
enzyme capacity enters HERE, exactly, not as a penalty
the box holds by construction, so there is no C_{box} loss term
b The constraint residuals hanging off v. One is exact by construction; the other four are soft, and none needs a binary variable.
Flux v [B, 4131]
Mass balance
$C_{bal} = \mathrm{mean}_i\, ([Sv]_i / \mathrm{sg}(\omega_i))^2$
sg = stopgrad; i runs over the 2593 independent rows of S
SOFT, scale-relative, squared
213 of the 2806 rows of S are redundant; the mean skips them
Second law
$C_{th} = \mathrm{mean}_j\, \mathrm{relu}(v_j\, \Delta_r G_j + \varepsilon)$
$/\ (|v_j|\, |\Delta_r G_j| + \varepsilon)$
a flux against its $\Delta_r G$ is penalized
SOFT hinge, no binaries
Thermo-Flux uses one binary variable per reaction; the hinge uses none
Enzyme capacity
folded into the box:
$ub_j \leftarrow \min(ub_j, k_{cat}$ capacity$)\, c_j$
no residual exists for it
EXACT
a bound (panel a, dynamic box), so v cannot violate it
Protein budget
$C_{bud} = \mathrm{relu}(\, P(v) / P_{avail} - 1\,)$
$P(v) = \Sigma_g\, MW_g\, E_g(v)$
$E_g(v) = \Sigma_j\, \sqrt{(v_j^2 + \delta)} / k_{cat,gj}$
SOFT hinge, one-sided
no reward for approaching P_{avail}; only exceeding it costs
Gibbs dissipation
$C_{diss} = \mathrm{relu}(\, g_{diss} / g_{lim} - 1\,)$
$g_{diss} = \Sigma_{j \in \mathrm{exchange}}\, \Delta_r G_j\, v_j$
g_{lim} = 3700 J gDW^{-1} h^{-1}
SOFT hinge, one-sided
g_{lim} from Niebel 2019
Thermodynamic decomposition of $\Delta_r G_j$, the driving force used by the second law and by dissipation
$\Delta_r G_j$ = $\Delta_r G'^0_j$ + RT $\Sigma_i\, S_{ij}$ ln c_i + $\Delta_r G^t_j$
tabulated: $\Delta_f G'^0$ known for 85.1% of metabolites
learned ln c_i, bounded to 0.1 μM to 10 mM; anchored where a measurement exists
transport term: recoverable for 441 transport reactions from the GEM's table; off by default
What rests on data, and how much of it is there
S 100%
$\Delta_f G'^0$ 85.1% of metabolites
MW 100%
k_{cat} 4.0%: 148 of 3728 catalytic units resolved from the Open Enzyme Database
conc. anchors 6.8% of metabolites
exact / structural
soft, squared, scale-relative
soft, one-sided hinge
data-backed quantity
learned
default, or a gap in the data